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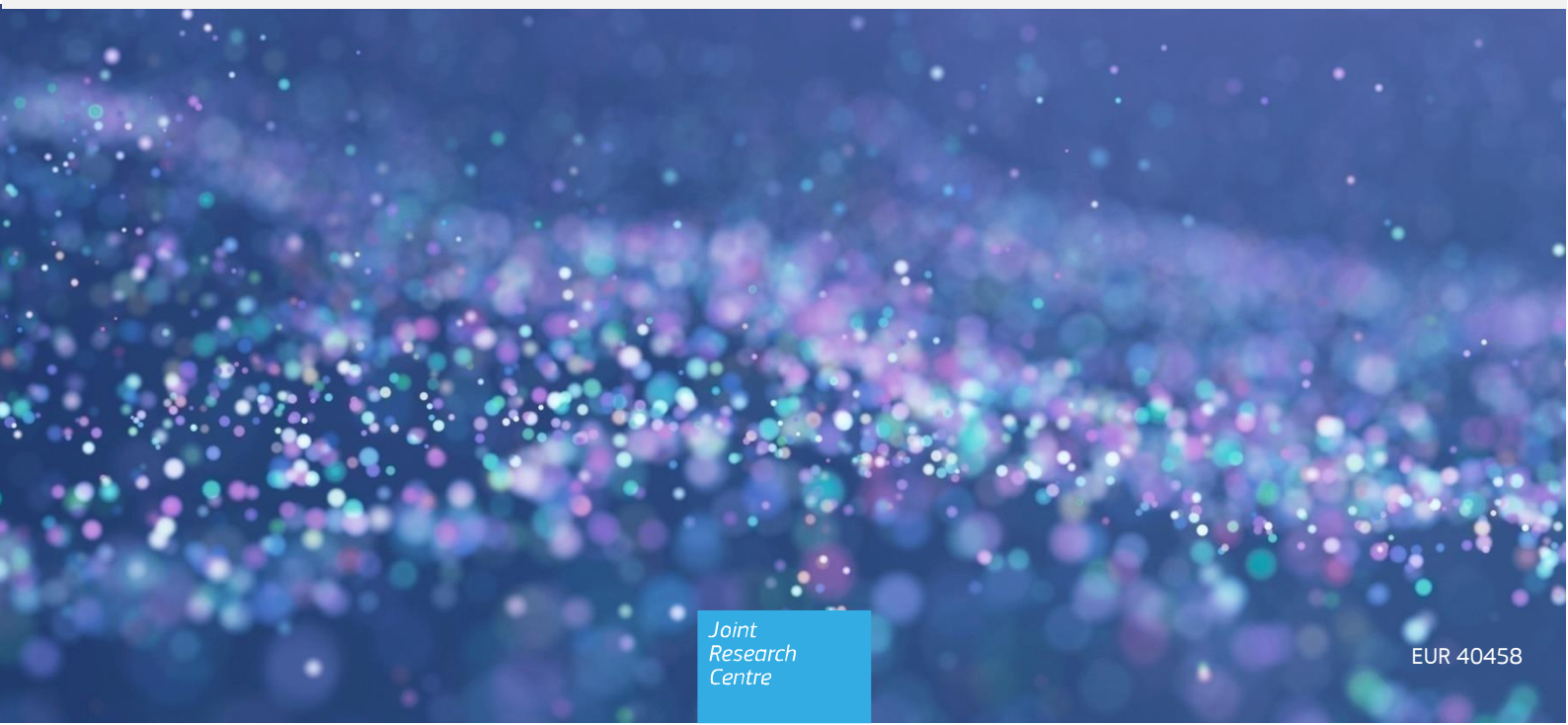



# AI-driven assessment of uncertainty and quality terminology in Copernicus data to enable policy uptake

*Application to the Copernicus Land Monitoring System*

Pierre, G., Urraca Valle, R.

2025




EUR 40458

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## Abstract

Copernicus data needs to be fit for legal applications to enhance its use in policymaking. Quality must be quantified and traceable to international standards. While various pixel-level quality indicators exist, only uncertainties provide the reliability required for legal context. This study aims to evaluate the uncertainty characterization and the consistency of quality nomenclature across Copernicus. We developed an AI-driven methodology to identify and classify quality layers—defined as information on observation count, quality flags, data spread, classification probability, errors, and uncertainty. We applied the methodology to the Copernicus Land Monitoring Service (CLMS), analysing 94 datasets and 757 layers. The main barrier was the lack of programmatic metadata access, prompting recommendations to enable similar studies across Copernicus.

We identified 189 quality layers across 61 datasets; 14 datasets reported uncertainties, but only 8 followed metrological guidelines. These could support use cases demonstrating how propagating uncertainties to policy indicators can enhance the uptake of Copernicus data in policymaking. We found inconsistent quality nomenclature in CLMS, with terms such as uncertainty, error, and standard deviation used interchangeably, and generic labels like flag or confidence varying in meaning across datasets. Copernicus services should standardize both the content and terminology of quality layers. A better definition of the uncertainty budgets and coverage factors is also needed.

## Acknowledgements

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***Authors***

Gaia Pierre, Ruben Urraca Valle

# 1 Introduction

Copernicus, the flagship Earth Observation (EO) programme of the European Union, offers open-access data to continuously monitor the Earth and its environment (Copernicus, 2020). A primary objective of Copernicus is to provide accurate information to inform policy makers, supporting the entire policy cycle. EO data can be used for policy anticipation, as reference data in impact assessments; for policy implementation, to monitor, report and verify the correct implementation of policies; and for policy evaluation, providing independent data to support ex-post assessments.

Policy officers are particularly keen on integrating EO data into policy implementation. Using EO data to monitor policies, instead of requesting it to public administrations or citizens, would significantly reduce the administrative reporting burden – a key priority of the current European Commission (European Commission, 2023). Despite the strong interest, the uptake of EO data in policy implementation is complex due to potential legal implications. When non-compliance with regulations arises, European institutions may initiate legal procedures, such as infringement procedures. Therefore, EO data needs to be fit for legal applications. It is not enough to develop EO datasets at the spatial and temporal resolution required. The quality of EO-based indicators needs to be quantified and traceable to internationally agreed standards (Widlowski, 2015).

Currently, the quality of Copernicus products is evaluated operationally through validation reports that use traditional validation methods such as comparison against ground stations or product inter-comparison to calculate error metrics (Loew et al., 2017). Then, these metrics are compared against the products requirements. While these metrics provide a general overview of product quality, they often fall short in quantifying the quality of EO-based policy indicators or other downstream applications. The quality of an EO-based indicator depends on aspects such as the spatial area evaluated, the temporal period, or the spatiotemporal resolution of the indicator – aspects not captured by traditional validation metrics. To address this, the quality of each individual observation is needed.

Uncertainties are the only quality indicator that provide the confidence needed for legal applications. Uncertainty methods and terminology are defined by the Joint Committee for Guides in Metrology (JCGM), and international body in charge of publishing the Guide of the Expression of Uncertainty in Measurement (GUM) (JCGM, 2023a) and the International Vocabulary of Metrology (VIMs) (JCGM, 2023b). While metrology has a long history of application to traditional instruments used in legal contexts (e.g., in-situ measurements), applying metrology principles to satellite-borne sensors is not trivial. Different methods have been developed in European Union (EU) research projects such as FIDUCEO (FIDUCEO, 2019), GAIA-CLIM (GAIA-CLIM, 2020) and QA4ECV (QA4ECV, 2014), leading to international frameworks such as the Quality Assurance Framework for Earth Observation (QA4EO) (European Space Agency, 2025). As a result, significant progress has been made in uncertainty characterization of ground segment data (Level 1, Level 2) in recent years (BIPM, 2025), particularly in the field of optical quantitative variables with initiatives such as the Optical Mission Performance Cluster (OPT MPC)(European Space Agency, 2015). Despite this, few Copernicus service products (Level 3, Level 4) currently provide uncertainties. Moreover, the use of uncertainty terminology, and quality nomenclature in general, remains inconsistent across and within EO services (BIPM, 2025).

This project aims to evaluate (1) how many Copernicus datasets provide pixel-level quality indicators, (2) the type of quality indicators provided, and (3) the consistency of the quality nomenclature across datasets. Our primary interest lies in identifying how many datasets provide

uncertainties and assessing the consistency of their uncertainty nomenclature with established standards.

To do this, we develop a three-step methodology that (i) retrieves the list of datasets and layers of a Copernicus services, (ii) extracts the information about the layer content, and (iii) identifies and classifies quality layers. We apply this methodology as a pilot case to the Copernicus Land Monitoring Service (CLMS), highlighting challenges that may hinder the methodology's broader application to all Copernicus services.

# 2 Copernicus Land Monitoring Service (CLMS)

The Copernicus Land Monitoring Service (CLMS) is one of the six thematic Copernicus services. It provides geographical information on Land Cover, and it changes, land use, ground motion, vegetation state, water cycle and earth surface energy variables for both Europe and the entire globe (Copernicus, 2018).

The service is divided into 6 thematic components:

Full-coverage Land Cover and Use

- Land Cover & Use in Priority Areas
- Bio-geophysical variables
- Ground Motion Data
- Land Satellite Mosaics
- Reference and Validation Data

**Full-coverage Land Cover and Use**

Land cover classifications at various level of detail, both within the global and pan-European context. At global level, the land cover mapping follows the Food and Agriculture Organization's (FAO) modular-hierarchical Land Cover Classification System. At pan-European level, it is complemented by detailed layers on land cover characteristics, such as imperviousness, forests, grassland, water and wetness and small woody features (Copernicus, 2025a) .

Main data format: The land cover data is available in raster format, of 10 or 100m spatial resolution over the time span from 2015 to 2020. The information contained is derived from various Earth Observation sources, mainly from Sentinel-2 but also from Sentinel-1, AgERA5, Landsat, DEIMOS-2, Pleiades 1A, Pleiades 1B, SPOT 6.

**Land Cover & Use in Priority Areas**

Tailored and more detailed land cover, land use and crop type information including its changes in specific areas of interest globally and in Europe, often called "hotspots". Hotspots in the context of our service are areas prone to environmental change such as urban environments, riparian areas, coastal zones, and key landscapes for conservation (Copernicus, 2025a).

Main data format: The high-resolution land cover and change data for environmentally sensitives areas is presented as a vector format, showing conservation borders but also a raster satellite image (30m).

**Bio-geophysical variables**

Qualified products on the status and evolution of the land surface globally and in Europe. Temperature and Reflectance, Snow, Soil Moisture, Water Bodies, Vegetation... (Copernicus, 2025a):

Main data format: This subcomponent stores products both under raster and vector, at various spatial resolution and time span.

### Ground Motion Data

Ground motion data, such as ground movements such as landslides, subsidence, and deformation, are provided across Copernicus participating countries by the European Ground Motion Service (Copernicus, 2025c).

Main data format: These files are provided under a vector format, with data derived from Sentinel-1.

### Land Satellite Mosaics

High-resolution satellite image from Sentinel-2 Global Mosaic (S2GM) production at global level as well as from contributing missions covering the territory of Europe (Copernicus, 2025d).

### Reference and Validation Data

Reference datasets providing homogeneous pan-European coverage of some key geospatial themes, such as hydrography, and in situ measurements collected at global scale to validate some of the bio-geophysical variables of the CLMS. The reference and validation data subcomponent is divided in 2 main services: Global Ground-based Observations for Validation Service and EU-Hydro (Copernicus, 2025a).

Main data format: the Global Ground-based Observations for Validation (GBOV) Service provides high quality in-situ measurements at global scale, in raster format. The EU-Hydro service is composed by vector files of the river network at the European scale.

# 3 Methods

## 3.1 Nomenclature

Terms such as dataset, layer, and quality layer are used extensively throughout the report. The lack of standardized international glossaries in the EO domain often leads to inconsistent usage of these terms. To ensure clarity and coherence within this report, the following specific definitions will apply:

**Dataset**

- Definition: Collection of data that shares the same processing chain (input data, retrieval algorithm), have the same spatiotemporal resolution and are released in a common data format. A dataset can contain information about one or several biophysical variables.
- Other names: product.
- Source: implicit definition given by CLMS. The concept of dataset is rarely defined explicitly and generally used interchangeably with product. However, the definition of dataset is usually given implicitly by EO services. In CLMS website, a "CLMS dataset catalogue" exists, so each entry of the catalogue can be considered as a dataset. WEkEO and CDSE also define a concept of CLMS dataset equivalent to that of the CLMS website.

**Layer**

- Definition: An array of values of the same type. A dataset can contain one or more layers.
- Other names: element or variable.
- Source: Environmental Systems Research Institute (ESRI, 2025)

**Quality indicator**

- Definition: A quality indicator shall provide sufficient Information to allow all users to readily evaluate the "fitness for purpose" of the Data or derived product. A Quality Indicator may be a number, set of numbers, graph, uncertainty budget, or a simple "flag".
- Source: Committee on Earth Observation Satellites (CEOS), QA4EO (QA4EO, 2020)

**Quality layer**

- Definition: A quality indicator providing information at layer level (for each observation).

**Quality layer classification**

Based on the previous definitions, we propose the following classification of quality layers

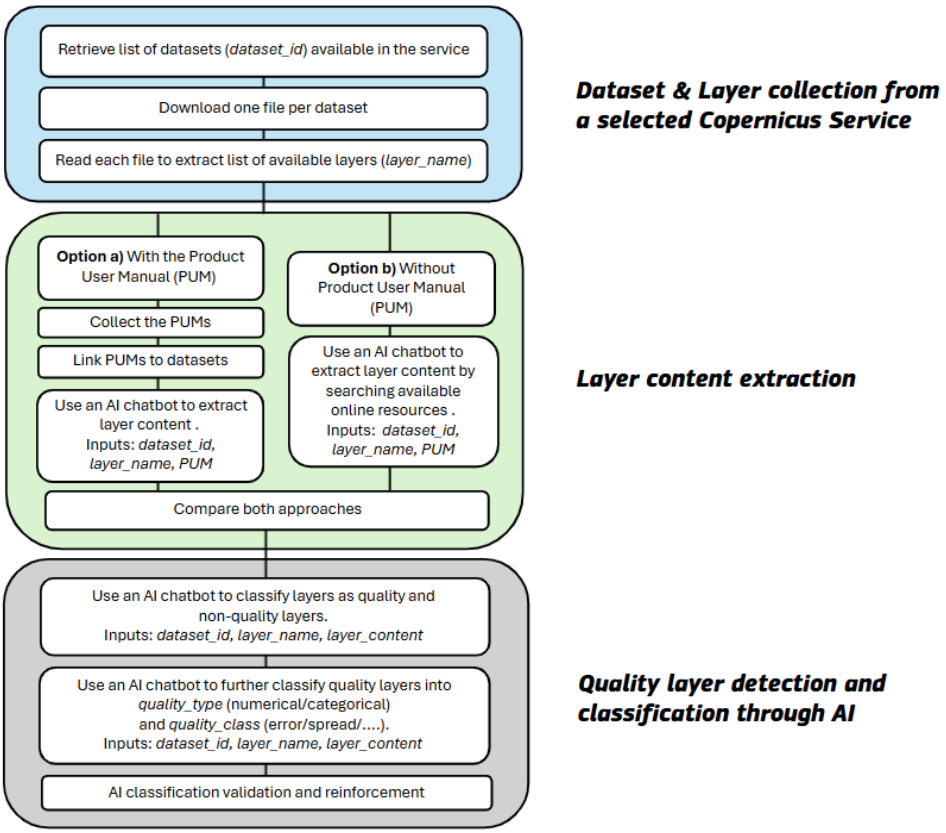
- **Quality type**: classification based on the data type of the quality layer into two categories: quantitative and qualitative.
- **Quality class**: classification based on the content of the quality layer into the following categories.
    - Number of observations. A layer that provides a quantitative quality indicator based on the number of observations used to make the retrieval.

- Spread: A layer that provides a quantitative quality indicator based on the spread of the observations (e.g., standard deviation) used to make the retrieval.
- Probability. A layer that provides a quantitative quality indicator based on the classification probability.
- Error. A layer that provides a quantitative quality indicator based on an error metric (e.g., Root Mean Square Error (RMSE)). An error metric measures the difference between the EO retrieval and a reference dataset. This error can apply to the entire retrieval or a specific part of the algorithm. Errors can contain both systematic and random components. The random component averages over multiple measurements, while the systematic error or bias doesn't (Niro et al., 2021).
- Uncertainty. A layer that provides a quantitative quality indicator based on uncertainty. Uncertainty is a parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand (JCGM, 2008)
- Flag: Any layer that provides a qualitative quality indicator.

## 3.2 Three-step methodology

To detect the presence of quality indicators within the CLMS datasets, a structured approach with three key stages was developed (**Figure 1**). The first step involved collecting a list of datasets (and layers) existing in the CLMS portfolio. The second step entailed extracting relevant information about the content of these datasets and layers. In the third phase, quality layers were detected and classified based on the information recovered in the second step using an Artificial Intelligence (AI) chatbot.

**Figure 1.** Flowchart of the methodology.




*Source: JRC*

### 3.2.1 CLMS dataset and layer collection.

The initial step of our project was to collect a comprehensive list of datasets provided by the CLMS. However, a significant challenge was encountered from the start: the CLMS webpage does not offer a centralized list of all available datasets. Instead, the datasets are scattered across its various subcomponents listed in Section 2, requiring manual navigation to locate and access them individually. The lack of a unified list led us to create our own inventory by compiling datasets from each subcomponent individually. Once the list was compiled, a representative sample file was downloaded from each dataset, but the absence of a straightforward list of datasets proved to be a major obstacle in the early stages of the project.

Another key objective of this project was to automate the download of files, to reduce the time and effort required for manual data retrieval. However, limitations in the CLMS website restricted bulk retrieval of the files, making it impractical to use as the centralized access point. As a result, different methods were used for each subcomponent, tailored to their specific storage locations.

We organized the CLMS products into the following categories, based on their data dissemination methods: global bio-geophysical variables & land cover, Global Hotspot Monitoring, S2GM, and European datasets. Each subcomponent provides files in raster formats at varying resolutions (e.g., 10m, 300m, 1km, 5km) or vector formats. These datasets also differ in temporal coverage, ranging from datasets spanning multiple decades to short-term, high frequency updates that capture seasonal trends.

As mentioned in Section 2, each subcomponent has different data access points, therefore the datasets were retrieved from multiple web interfaces. For the Bio-geophysical variables and Global Land Cover, the collection point was the manifest file. A script was developed to download the most recent file of each product. European datasets were also collected programmatically from WeKeo using the HDA python API. For the other sub-components, the recovering of the datasets was done manually. The Global Hotspots files were sourced via the Copernicus Hot Spot Land Cover Change Explorer webpage, and the S2GM from the Sentinel-2 Global Mosaic Hub (Copernicus, 2025e).

**Table 1.** CLMS datasets sources and access methods

| **Group of datasets** | **Source** | **Type of download** |
|---|---|---|
| Global bio-geophysical variables & land cover | manifest file | Programmatical |
| Hotspot monitoring | Copernicus Hot Spot Land Cover Change Explorer | Manual |
| S2GM | S2GM Hub | Manual |
| European datasets | WEkEO | Programmatical (HDA API) |

*Source: JRC*

Subsequently to their collection, every file was read to extract the list of layers. The files were not formatted consistently; hence the application of different python packages was required during the programmatic reading of their content. The file formats were encountered are detailed in **Table 2**.

**Table 2.** Python packages used to read the different CLMS data formats.

| **File format** | **Python package used** |
|---|---|
| Netcdf | Xarray |
| Tiff | Rioxarray or Rasterio |
| Json | Pandas |
| Geojson | Geopandas |
| Shapefile | Geopandas |
| Geodatabase | Geopandas |
| Geopackage | Geopandas |

*Source: JRC*

In addition of the format of the files being inconsistent, the internal structures of the datasets displayed differences, with information about layers stored in separate places: layer attribute, file attribute, independent file, no information. When the location of the layer information was identified, a script was generated to extract it, adapting the algorithm to the format and structure of each file. The names of all CLMS files and their layers were stored under a unique CSV file.

### 3.2.2 Layer content extraction

Once the list of datasets and layers was available, the second step consisted of extracting information about the layer content. Two approaches were implemented.

The first approach consisted of extracting this information from the associated Product User Manuals (PUM). Indeed, the PUMs contain the descriptions of the data and its layers, which would later be used to classify the type of layers present in the file. To facilitate the retrieval of PUMs, a custom script was elaborated to extract these documents from the CLMS website. The availability of PUMs was inconsistent throughout the CLMS portfolio. Normally, one PUM was available for each dataset. However, for European products, PUMs generally described multiple datasets, while exceptionally some datasets (e.g., Global Hotspot Monitoring) did not have a PUM. Moreover, linking the PUMs with their corresponding dataset was not trivial, due to the lack of a unique dataset identified. To solve this, we linked the datasets with the corresponding PUM manually.

Upon the creation of this comprehensive document combining all the datasets from the CLMS and their associated layer names, the next step was the extraction of layers description using an AI chatbot. To do this, a prompt to instruct the AI chatbot to recover the descriptions of the layers from the associated PUMs pdf, as well as the range of possible values, was written (**Box 1**).

**Box 1.** Prompt to extract the layer content when the Product User Manual (PUM) is available.

*For each layer of the specified Copernicus Land Monitoring Service dataset, extract the following information from the PDF manual:*

*- layer_description: the description of the layer -*
*- layer_range: the range of possible values of the layer*
*- layer_values: for categorical layers, list of potential categories.*

*dataset = %s*
*list of layers = %s*
*PDF manual = %s*

Due to the abovementioned problems to extract PUMs and linked them with their corresponding dataset, a second approach was tested. The chatbot was requested to provide an additional layer description based on the resources available online, without providing the PUMs (**Box 2**).

**Box 2.** Prompt to extract the layer content when the Product User Manual (PUM) is not available.

*For each layer of the specified Copernicus Land Monitoring Service dataset, provide:*

*- layer_description: the description of the layer*
*- layer_range: the range of possible values of the layer*
*- layer_values: for categorical layers, list of potential categories.*

*dataset = %s*
*list of layers = %s*

### 3.2.3 Quality layer detection and classification

Following the parsing of descriptions, the AI chatbot was instructed to categorized layers as either quality layers or non-quality layers based on keyword analysis (**Box 3**). Words commonly associated with data quality, such as "number of observations," "flags," "quality flags," "errors," "confidence," "standard deviation," and "uncertainty", were flagged. If the layer descriptions, both the one retrieved on the PUM and the one generated by the AI model, contained one of the flagged words, it was categorized as quality layer (True). If no relevant keyword were present in the description, the layer was labelled as non-quality layer (False).

**Box 3.** Prompt to detect quality layers

*Given a table with the name of a Copernicus dataset layer (layer_name) and the description of the layer (layer_description'), add the following columns*

*- quality_layer: a boolean equal to True is the layer contains a quality indicator such as number of observations, flags, quality flags, errors, confidence, standard deviation, and uncertainty*

*table = %s*

To go further in the understanding of the quality layers provided, a second query was made to categorize them by 'type,' differentiating between 'categorical' and 'numerical' indicators. A final classification was requested, to assign a specific 'quality class' based on the keywords highlighted in the initial prompt (**Box 4**).

**Box 4.** Prompt to classify the layers

*Given a table with the name of a Copernicus quality layer (layer_name) and the description of the layer (layer_description'), add the following columns*

*- quality type: a string indicating the data type of the quality layer, either 'categorical' or 'numerical'.*
*- quality_class: string assigned from the following predefined categories, based on the layer_description:*
  *- n_observation: layers reporting quality based on the number of observations.*
  *- flag: layers providing qualitative flags as quality indicators.*
  *- spread: layers where quality is based on the data's spread (e.g., standard deviation).*
  *- probability: layers providing classifier's probability.*
  *- error: layers providing error values (e.g., Root Mean Square Error - RMSE).*
  *- uncertainty: layers providing uncertainty values.*

*table = %s*

Based on the results obtained after the first script-based classifications, the prompts were adapted to correct small incoherences. This step is known as reinforced training. After a few trials and errors, modifications were made to the prompt, to enhance the AI model performance and optimize its calibration to reduce misclassifications.

## 3.3 Implementation

All the data computations and scripts for this project were performed within the Joint Research Centre (JRC) shared internal ecosystem, the Big Data Analysis Platform (BDAP) (Soille et al., 2023), an open-source software landscape. The primary data analytics services used were the data science desktop (JEO-desk) and the integrated development environment (IDE) PyCharm.

To handle, format and analyse our data, the following packages from the programming language Python (version 3.10) were used:

- Data collection and web scraping: requests, bs4, urljoin
- Data manipulation: pandas, numpy, xarray, geopandas, rasterio, rioxarray, json
- Data visualization: matplotlib
- Files and systems operations: time, os, sys, signal, glob

- AI chatbot implementation: google-genai, tika

For layer content extraction and classification, the Google AI chatbot *Gemini 2.5 Flash Preview (05-20)* was employed via the *google-genai Python* API. This model has a throughput of 10 requests per minute and a daily limit of 500 requests. The activation of an API key was necessary to allow the interaction between the AI model and the local environment, and to enable the automated processing of the data.

# 4 Results

## 4.1 Validation of the methodology

**Comparing the extraction of layer content with and without PUMs**

The first part of the validation consisted of comparing the two approaches used to extract the layer content with the AI chatbot. The first approach required collecting the PUMs and matching them with the datasets. The second one consisted of relying on the resources available online to get the layer content. The second approach is much faster and easily applicable to other Copernicus service, but its accuracy needs to be verified.

The main differences found were:

- Layer descriptions generated without the PUM can be incorrect or mislabelled. For example, without its PUM the Lake Ice Extent (LIE) dataset is described as Leaf Inclination Angle by Gemini.
- Layer descriptions without PUM are generally shorter than those with PUM, providing less information to detect and classify quality layers. Often the AI chatbot provides guesses correctly the general content of the layer but it is not able to provide a detailed description on how this indicator was calculated. For instance, the content provided for the layer “nobs” was a generic “number of observations”, without indicating how exactly what those observations represent.
- Sometimes the two approaches give contradictory contents: for instance, the same variable can be described as "Uncertainty on fAPAR 1km" in the PUM version, but "fAPAR error" in the version without the PUM. In those cases, the description provided by the PUM version is normally the most correct.
- Descriptions without PUMs can offer clearer explanations of what the layer represents, providing additional context that goes beyond the title.
- The descriptions without the PUM often include full expansions of acronyms, making them easier to understand for average users.

Based on the previous results, we decided to detect and classify quality layers based on layer descriptions obtained with the PUMs, due to their higher accuracy. Looking to the future, the development of AI chatbot specialized for Copernicus services could help to improve the accuracy of this type of prompts. In this sense, the recently released Copernicus Observia AI is a first step, but could not be used in this project due to the lack of API access.

**Overall accuracy of the AI chatbot classification of quality layers**

The AI chatbot successfully generated descriptions and classified all the layers. However, a few inaccuracies in the classification of some layers were detected. There are two main types of errors: false positives, layers erroneously classified as quality layer, and false negatives, quality layers overlooked by the model. While no false negatives were observed, the false positives described in **Table 3** were found.

**Table 3.** List of false positives (layers erroneously classified as quality layers)

| Dataset | Layer | Description |
|---|---|---|
| LIE-NH_NHEMI_SLSTR_V1.3.0, LIE250_Baltic_MODIS_V1.2.2 | LIE | LIE (lake ice extent) is the main output layer, classifying pixels into ice, open water, cloud, land and sea |
| SWI-STATIC-DGG_GLOBE_ASCAT_V3.0.1 | land_flag | Indicates if location is over land or not. Set to 0 if over water and 1 if over land. |
| SWI-STATIC-TC_GLOBE_GTOPO30_V1.1.1 | Topographic_complexity | Topographic Complexity is a measure of the variation in elevation within a grid cell; calculated from the GTOPO30 digital elevation model. It ranges from 0 to 100; with higher values indicating more complex terrain. This information can be used to mask areas where soil moisture retrieval might be difficult due to mountainous terrain. |
| SWI1km_CEURO_SCATSAR_V1.0.2, SWI_GLOBE_ASCAT_V3.2.1 | SSF | Surface state flag<br>255: Could not be determined; 0: Unknown; 1: Unfrozen; 2: Frozen; 3: Temporary Water on the surface |
| S2GM SIAC | cloud_confidence | Cloud confidence. Ranging from 0 for high confidence clear sky to 100 for high confidence cloudy. |
| EO:EEA:DAT:RIPARIANZONES___RZ_change | Urban atlas 2012 flag, urban atlas 2018 flag | Urban Atlas code associated with the riparian zone for the year 2012 or 2018 |

*Source: JRC*

In general, the false positives were layers describing the state of land or clouds during the retrieval. While these layers are not an explicit quality indicator, they could impact the quality of the retrieval, explaining why they were considered as quality indicators by the AI chatbot. On the chatbot side, the main error was the classification of urban atlas flags in the Riparian Zones dataset as quality layer, which was likely due to the presence of the term “flag” in the layer name.

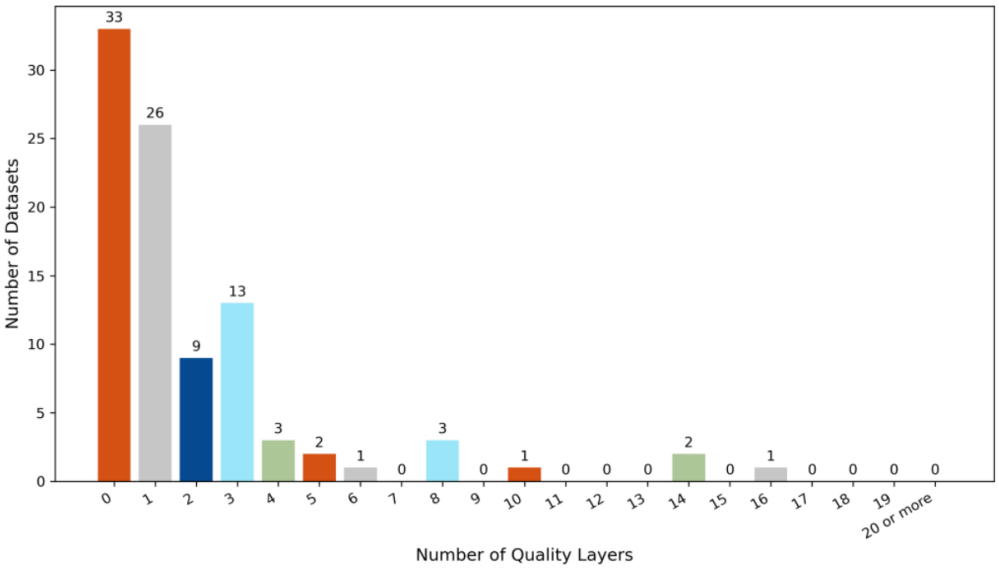
Some errors were also detected in the classification of quality layers into the correct quality class. However, these errors are not always related to the AI chatbot accuracy, but to the nomenclature used by the data providers to name the layers and describe it. These errors are further discussed in Section 4.2.2

## 4.2 CLMS quality layers

### 4.2.1 Descriptive statistics

A total of 94 datasets and 757 layers were analysed from the CLMS portfolio. Following the AI-based classification methodology, 189 layers (25% of the total) were classified as quality layers. Besides, 61 datasets (65% of the total) contained at least one quality layer. Most datasets contain one (26 datasets), two (9 datasets) and three (13 datasets) quality layers, but some of them can contain up to 14 or 16 quality layers (**Figure 2**).

**Figure 2.** Number of quality layers per CLMS dataset.




*Source: JRC*

**Table 4** classifies CLMS quality layers by data type, categorical or numerical. Some datasets contained both categorical and numerical quality layers and have therefore been doubled in this table. The breakdown of the datasets reveals that 25 datasets contained only categorical quality layers, 18 had only numerical quality layers, and 18 datasets provided both categorical and numerical quality indicators.

**Table 4.** Summary of CLMS quality layers by data type, categorical or numerical.

| | **Number of datasets** | **Number of layers** |
|---|---|---|
| Categorical | 43 (46%) | 77 (10%) |
| Numerical | 36 (38%) | 112 (15%) |

*Source: JRC*

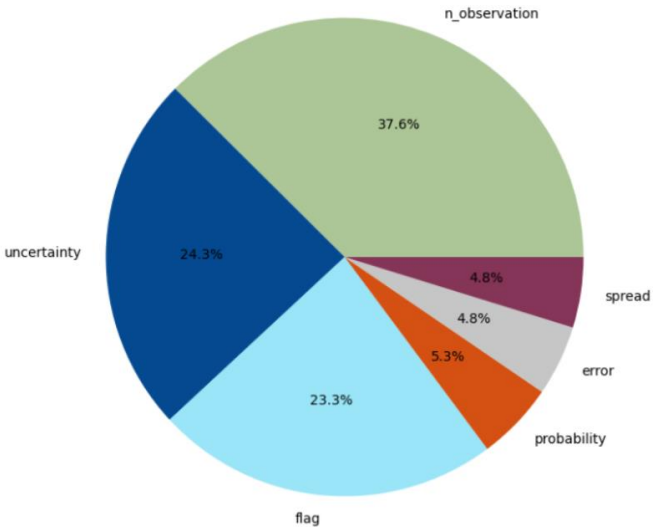
**Table 5** and **Figure 3** classify the quality layers by quality class. As in **Table 4**, some datasets some datasets carry multiple layers classified under different quality classes and were therefore counted separately in each relevant class. 'Number of observations' is the most common quality metric (over one third all quality layers), followed by 'uncertainty' and 'quality flag'. Metrics like 'error' and 'spread' and 'probability' are relatively rare, representing less than 10% of the total quality layers. Only 14 datasets contained the quality indicator 'uncertainty', meaning that 85% of the CLMS datasets lack a crucial indicator for assessing the quality of the data.

**Table 5.** Summary of CLMS quality layers by quality class

| | Number of datasets | Number of layers |
|---|---|---|
| Number of observations | 31 (33%) | 71 (9%) |
| Spread | 9 (10%) | 9 (0.1%) |
| Flag | 38 (40%) | 44 (6%) |
| Error | 9 (10%) | 9 (0.1%) |
| Uncertainty | 14 (15%) | 46 (6%) |
| Probability | 5 (5%) | 10 (0.2%) |

*Source: JRC*

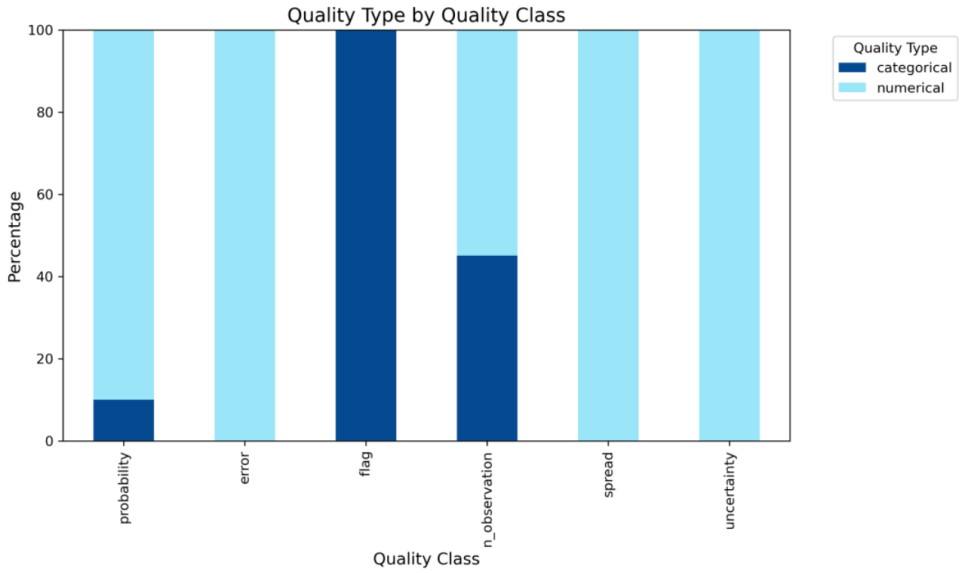
**Figure 3.** Distribution of CLMS quality layers by quality class.




*Source: JRC*

In several cases, layers were classified under the same quality class but had different data types (**Figure 4**). One example is the 'probability' class, which was classified as a numerical quality metric 90% of the time and as a categorical metric 10% of the time. Similarly, half of the 'number of observations' were counted as a numerical indicator, and the other half as a categorical indicator. In both cases, the reason is that some datasets provide custom quality categories based on the number of observations or the classification probability, instead of providing the raw numeric values.

**Figure 4**. Data type (categorical or numerical) of the different quality classes.




*Source: JRC*

### 4.2.2 Consistency of quality nomenclature

**Number of observations**

Content consistency

Good consistency between the content provided by all these layers. All of them provide information about the number of observations used for the retrieval, or for the spatiotemporal aggregation of the output variable (e.g., observations used within a composite period). Most datasets provide numerical layers, with either absolute (total number of observations) or relative (percentage of valid observations) information. The exception is global SWI, which provide custom categories (quality flags) based on the number of observations.

Nomenclature consistency

The layers are named with a wide variety of acronyms: n_, NOBS, FRAC_VALID_OBS, num_obs, n_obs, NOBS, VOBS, valid_observations, stats_valid. The least consistent layer names are DataDensityIndicator (PROBAV_LC100_global_v3.0.1), DataScore Layer (CLC-PLUS), and QFLAG (SWI10_GLOBE, SWI1km_CEURO). Due to the lack of consistency, the latter layer was originally classified by the AI chatbot in the “flag” class.

**Data spread**

Content consistency

Good content consistency, with all layers providing metrics of the data spread. The most common one is the standard deviation of the observations used for the retrieval. S2GM temporal composites, which provides as data spread the sum of the distances between each observation and the medoid normalized by the total number of observations. HRL TCF (tree cover & forests) provides the standard deviation of an ensemble of models used to estimate tree cover density.

Nomenclature consistency

The layers providing standard deviation used different: standard_deviation, sigma, stdev, medoid_quality_band. The layer names used by S2GM and HRL Tree Cover & Forests do not reflect the type of quality information provided (data spread). S2GM layer is "mediod_quality_band" and the HRL Tree Cover & Forests is "Tree Cover Density Confidence Layer".

**Classification probability**

Content consistency

The classification probability is a specific case of a data spread metric for classifiers. Most layers provide classification probability of the class as an estimate of the quality of the categorical product. (Land cover 100m, HRL Grasslands). In some cases, the difference between the probability of first and second class is provided.

Nomenclature consistency

The name of the layers does not reflect the type of information provided: *discrete-classification-proba* (land cover 100m), *confidence layer* (HRL Grasslands, HRL Croplands, HRL Tree Cover & Forests)

**Flag**

Content consistency

This is the largest and most heterogeneous class. All the layers provide qualitative quality information based on a heterogeneous group of quality drivers. The drivers used include the presence of missing data, gap filled data, cloud shadows, clouds or water. Expert flags, the type of input sensor or the specific configuration of the retrieval algorithm are also used as quality drivers.

Nomenclature consistency

The name of these layers includes terms such as QFLAG, quality_level, flag, QUAL, QC. The number of quality classes normally oscillates around 3-4 but can be up to 10 (HR Vegetation & Phenology Parameters). The labels of the quality classes are heterogeneous and different for each dataset. Some examples are:

- *0: Good data; 1: dubious data; 2: not available; 3: Fill value*
- *0: Good data; 1: Acceptable data; 2: Bad data; 3: No data*
- *0: Good quality; 1: Acceptable quality; 2: Bad quality; 3: Not produced*
- *0: Good; 1: Marginal; 2: Poor; 3: Reserved; 4: Not available*

**Error**

Content consistency

The type of information presented by "error" quality layers is highly heterogeneous, due to the inconsistent use of the term error by the data providers. Despite all these layers include the term error in their name and description, only the correlation layer provides a kind of error, understood as the difference between satellite retrieval and an external reference.

- RMSE (LAI, FAPAR, FCOVER 300 m): The RMSE for each variable are calculated as the root mean square difference between the instantaneous estimates in the compositing window and the final 10-day product value. Thus, it is not an error, but a data spread metric.
- Error bar (LST_GLOBE_GEO): This layer provides an error budget including the following error sources: water vapour, surface emissivity uncertainty, sensor noise, viewing geometry. The different error sources are characterized individually and then propagated mathematically. Thus, this metric is closer to metrological uncertainty than to an error metric.
- Noise (SSM_1km): standard deviation of the input data. This layer should be part of the number of observations class.
- Pearson correlation (SWI_STATIC). Correlation between CLMS soil moisture and GLDAS NOAH soil moisture.

Note that in this case the AI chatbot correctly classified these layers. The problem was on the incorrect usage of the term error in the datasets. We did not re-classify them to highlight the inconsistent use of the error term.

Nomenclature consistency

Terms used: *RMSE, error bar, correlation and noise.*

**Uncertainty**

Content consistency

This is the most relevant class for our study, so we analyse all the datasets individually:

- Global LAI, FPAR, FCOVER, 1km. All these datasets provide an *err* layer, defined as an uncertainty value in their PUMs. However, these layers contain RMSEs of the retrieval algorithm derived from the training database, not uncertainties.
- Global NDVI, 300m. The layer *NDVI_unc* provides the NDVI uncertainty calculated by propagating the uncertainty associated to TOC reflectance. The uncertainty is calculated by propagating TOC reflectance uncertainty into the NDVI equation. However, the ATBD does not describe which values are used to quantify the uncertainty of Sentinel-3 TOC reflectance.
- Lake surface water temperature (LSWT), 1km. This dataset provides a *LSWT_uncertainty* layer calculated with the algorithm developed to ESA CCI from MacCallum S. & Merchant C.J., (2013) and Merchant & SST CCI Team, (2017). The uncertainty sources considered are instrumental noise, random forward error model, and errors in prior state of the inversion model.

- Lake water quality (LWQ), 300m. This dataset provides uncertainty layers for water-leaving reflectance (*RW_relative_uncertainty, RW_relative_uncertainty_unbiased*), chlorophyll (*chla_uncertainty*) and total suspended matter (*tsm_uncertainty*). The algorithm used to characterized uncertainty is based on Liu et al., (2021), as developed for ESA CCI Lakes. Uncertainty is calculated based on the relation between satellite vs in situ residuals and Optical Water Type (OWT) similarity. The per-pixel uncertainty for water-leaving reflectance propagates uncertainty from regression fits between satellite and in situ observations, without discriminating OWTs. Therefore, this quality indicator is more an error metric than a measure of uncertainty.

- S2GM variables based on SIAR. The S2GM provides uncertainty layers for bi-directionally reflectance factors (BRF), Nadir BRDF-Adjusted Reflectance (NBAR), albedo, LAI and FAPAR. Uncertainties are calculated by assuming a Sentinel-2 L1 uncertainty of 5% and then propagating these uncertainties across the retrieval algorithms.

- Snow Water Equivalent (SWE), 5km. This dataset provides a layer *swe_var* that according to the ATBD provides the statistical uncertainty of the SWE layer. The SWE retrieval algorithm is as follows. The snow depth and effective snow grain size at SYNOP locations are interpolated spatially using kriging. Then, in each pixel a snow model is inverted to obtain SWE based on the interpolated grain size and the observed passive microwave data from satellites. The uncertainty reported by this algorithm includes both the effect of the systematic (bias) and statistical (random) error. Statistical error is determined through an adaptive dynamic error propagation approach during the kriging interpolation (Takala et al., 2011, Luojus et al., 2021). The systematic error is derived from a multi-year analysis of independent snow course observations from Russia and the Former Soviet Union. Therefore, the quality information provided by *swe_var* corresponds better to the data spread metric than to uncertainty.

- Water Level (WL). This dataset provides a layer called *associated_uncertainty* that represents the standard deviation from height of high frequency measurements used in the estimation. Therefore, this is a quality indicator reporting data spread, not uncertainty.

- HRL Croplands. This dataset *confidence* layers for most of the main cropland layers. As discussed in the probability class, the confidence layers related to crop classification contained information about the classification probability. On the contrary, the confidence layers related to temporal events (main crop, bare soil after, bare soil before, secondary crop duration, fallow land duration) provide uncertainty information. This uncertainty is calculated by propagating the uncertainty of the FAPAR, which is calculated using the CropSAR algorithm. The ATBD does not describe how the FAPAR uncertainty is calculated.

- HRL Grasslands. As for the cropland dataset, HRL Grasslands provide confidence layers for most of their layers. Only one of these layers can be considered an uncertainty, the *Grassland Mowing Events Confidence Layer*. As described in the Croplands dataset, this layer provides the uncertainty in days for moving events. However, contrary to HRL grasslands, it is not calculated by error propagation. The confidence value is based on an empirical formula that depends on the number of missing Sentinel-2 observations, and the likelihood of cloud contamination (proximity to cloud-cover pixels).

Nomenclature consistency

High level of inconsistency, with datasets mixing terms such as uncertainty, error, standard deviation or confidence. Sometimes one term is used in the layer name and other term in the layer description offered by the PUM.

### 4.2.3 CLMS datasets providing pixel-level uncertainties

Based on the previous analysis, we classified all the uncertainty layers in uncompliant (**Table 6**) and compliant (**Table 7**) with metrological guidelines. Layers deemed non-compliant claim to provide uncertainty information but instead report error metrics, data spread metrics, or generic quality flags. It is important to note that data spread metrics, such as the standard deviation, can be used to characterise uncertainties—specifically the so-called Type A uncertainty—when they refer to the same measurand. However, in the context of EO, spread metrics are often calculated using measurements taken over time (temporal standard deviation) or across space (spatial standard deviation). In such cases, the measurand of each observation is different, and the spread metric reflects both the uncertainty associated with each measurand and the natural spatial or temporal variability of the phenomenon itself.

**Table 6.** List of datasets providing uncertainty layers uncompliant with metrological guidelines.

| Dataset | Layers | AI quality class | True quality class |
|---|---|---|---|
| FAPAR 1km<br>LAI 1km<br>FCOVER 1km | *FAPAR_err*<br>*LAI_err*<br>*FCOVER_err* | Uncertainty | Error |
| LWQ 300m | *RW_relative_uncertainty*<br>*RW_relative_uncertainty_unbiased*<br>*Chla_uncertainty*<br>*Tsm_uncertainty* | Uncertainty | Error |
| SWE 5km | *Swe_var* | Uncertainty | Error & spread |
| WL | *Associated_uncertainty* | Uncertainty | Data spread |
| HR grasslands | *Confidence* | Uncertainty | Flag |

*Source: JRC*

Datasets in **Table 7** are compliant with metrological guidelines, and therefore those that could be used to develop pilot cases of propagating uncertainties to policy indicators. Note however that none of them are fully compliant with the metrological guidelines. All of them either exclude some uncertainty sources or make assumptions about the input uncertainty values, which can lead to over- or underestimation of the true uncertainties. Nevertheless, these datasets represent valuable first steps in the right direction.

**Table 7.** List of datasets providing uncertainty layers compliant with metrological guidelines

| Dataset | Layers | Note |
|---|---|---|
| NDVI 300m | *NDVI_unc* | The method used to quantify Sentinel-3 TOC uncertainty is unclear." |
| LSWT 1km | *LSWT_uncertainty* | Only some uncertainty sources are considered. |
| S2GM (SIAC atmospheric correction) | *Bx_unc*<br>*DHR_unc*<br>*BHR_unc*<br>*FAPAR_unc*<br>*LAI_unc* | Assume a 5% uncertainty for Sentinel-2 TOC. |
| LST 5km | *Errorbar_lst* | Reported as an error in the PUM despite providing an uncertainty budget |
| HR croplands | *Confidence* | The method used to quantify FAPAR uncertainty is unclear." |

*Source: JRC*

### 4.2.4 Summary and recommendations

**Regarding CLMS datasets providing uncertainties**

- Only 5 datasets (~5% of total CLMS portfolio) provide uncertainty layers compliant with metrological principles.
- All of them make assumptions regarding the uncertainty sources considered and their characterization, but they represent good steps in the right direction.
- PUMs and ATBDs should better describe the uncertainty budgets, include uncertainty trees (currently not included in any CLMS dataset) and describe all the assumptions made during uncertainty characterization.
- PUMs should clearly describe the coverage factor of the uncertainties (currently missing).
- All these datasets have benefited from research projects focused on the uncertainty characterization of EO measurements (e.g., ESA CCI End-to-End Uncertainty Budget, ESA HR Albedo), underscoring the importance of continued research to develop methodologies and tools for the uncertainty assessment of Copernicus datasets.
- These datasets could be used to deliver use cases of uncertainty propagation in policy indicators, to show the value of uncertainty-quantified indicators in applications with legal implications.

**Regarding all CLMS datasets**

- Several datasets claim to provide uncertainties that are not compliant with metrological guidelines. Data providers frequently mixed terms such as uncertainty, error and temporal standard deviation.
- Generic terms such as confidence or quality or flag, that mean different things in each dataset, shall be avoided. A good example is the generic term “confidence” used by HRL croplands and grasslands. It was used to name (1) classification probability, (2) custom quality flag based on missing values and cloud contamination and (3) uncertainty in days propagated from the FAPAR uncertainty.

- CLMS should work towards standardizing its quality indicators, both in nomenclature and content. The current shift toward a centralized contract for dataset validation offers an opportunity to advance this goal.

## 4.3 Expanding the analysis to all Copernicus services

### 4.3.1 Barriers

Most of the barriers to implement the methodology and to extend it to all the Copernicus services appear in the first step of the methodology – dataset and layer collection. Indeed, the need of this step in the methodology already proves the existence of a barrier. Ideally, the information needed to conduct this study (a list of the datasets and layers of the Copernicus services) should be accessible in a centralized database and accessible programmatically through a metadata catalogue. There are some different catalogues at Copernicus and service level, but none of them meet the requirements for a project requiring programmatic access to a comprehensive list of metadata parameters.

Existing centralized catalogues:

- Copernicus Services Catalogue (Copernicus, 2025b): Official Copernicus catalogue providing a comprehensive list of datasets from all Copernicus services. However, the catalogue is (1) not updated regularly, (2) does not include all Copernicus datasets, and (3) does not provide metadata information (only dataset name and description, with the link to the specific service).
- Centralized data platforms such as CDSE or WEkEO implement their own catalogues to organize the data and make it accessible. WEkEO appears as the best candidate providing access to data and metadata from all Copernicus services. However, it shares some of the same limitations as the previous catalogue. WEkEO does not include all Copernicus datasets (e.g., most of the Copernicus Emergency Monitoring Service (CEMS) data is missing). Despite providing a more comprehensive list of metadata, metadata changes across datasets are and still insufficient for many applications.

Service-level catalogues

- CLMS is in the process of on-boarding datasets to CDSE. This will make possible to access data and metadata throughout the CDSE SpatioTemporal Asset Catalog (STAC).
- The Copernicus Climate Change Service (C3S) and Copernicus Atmosphere Monitoring Service (CAMS) offer standard metadata access with STAC and Catalogue Service for the Web (CSW).
- Copernicus Marine Environment Monitoring Service (CMEMS) offers a specific python API.
- CEMS doesn't have a centralized data catalogue. Each CEMS component adopt a different strategy to disseminate their data and metadata.

**Barriers related to existing centralized and service-level metadata catalogues**

- Centralized data catalogues are not always updated regularly.

- Centralized data catalogues are incomplete. Some service-level catalogues are also incomplete. For instance, the CLMS dataset catalogue does not include S2GM, GBOV or Hot Spots Monitoring (HSM). In CMEMS, the catalogue is inexistent.
- Available catalogues do not provide a comprehensive and standardized list of metadata parameters.

Based on the previous limitations, the solution was to create ourselves the list of datasets and layers by downloading one file from each dataset.

**Barriers to automated identification of the list of datasets of a Copernicus service**

- The full list of datasets comprising a service is unclear in services without data or metadata catalogues (CEMS), or services with uncomplete data catalogues (CLMS).
- Inconsistent dataset definition across and within services. For instance, in CLMS each entry of the "dataset catalogue" could be in principle considered a dataset. Based on this assumption, we can observe some inconsistencies. For bio-geophysical variables, a dataset encompasses all temporal observations (e.g., "Lake ice extent 2024-present"). Conversely, for land cover, a distinct "dataset" entry is created for each annual update (e.g., "Land Cover 2017 (raster 100m)" or "CORINE Land Cover 2012").
- Lack of a standardized dataset identifiers (e.g., "dataset_id"). While some identifiers are generated when datasets are uploaded to data platforms like WEkEO or CDSE, these are rarely reflected in the primary dataset documentation or service catalogues. Also, the definition of these identifiers is not consistent across and within services, with identifiers of different datasets including different information. Moreover, sometimes the identifiers are not unique and change across platforms (e.g., the same C3S dataset has different identifiers in the C3S data store and in WEKEO).

**Barriers for automated data download**

- Not all the services offer direct cloud access to their datasets. This means that some datasets must be downloaded locally before they can be accessed (e.g., HSM in CLMS).
- To download the data, not all the services provide APIs to download the files programmatically. As with the catalogues, some of the existing APIs do not cover the whole portfolio. For instance, CDSE does not include S2GM, GBOV or Hotspot monitoring datasets. Tailor download scripts for specific parts of the services. In the worst scenario, the download must be done manually (CEMS drought data, CLMS HSM).

**Barriers for automated data reading**

- Lack of standardized data formats. For instance, CLMS provides raster datasets in NetCDF, TIFF, Shapefile, and GeoTIFF. Similarly, vector data is available in Shapefile, GeoJSON, Geodatabase, and Geopackage formats. Each format necessitates a different approach (e.g., python packages) when developing scripts for data reading the data
- Lack of standardized data structure. Even within a specific file format, the internal organization of the data can vary leading to a further customization of the reading scripts.

- Inconsistent distribution of the metadata. Linked to the previous point. For this project, metadata about layer content in CLMS datasets was (a) available as an attribute of a layer, (b) available as an attribute of the dataset, (c) available as a separate metadata file, (d) unavailable.

**Barriers to automate information extraction from Product User Manuals (PUMs)**

- Some products do not have PUMs or ATBDs. In CLMS, this was the case of HSM.
- The absence of dataset identifiers (dataset_id) prevents programmatic linking between the Product User Manuals (PUMs) and Algorithm Theoretical Basis Documents (ATBDs) with their corresponding datasets.
- Inconsistency relationship between PUMs/ATBD and datasets. In CLMS, most JRC-produced datasets have one PUM per dataset. However, European Environment Agency (EEA) based produced datasets have one PUM per contract, including the description of several datasets in the same document.

### 4.3.2 **Recommendations:**

**On metadata catalogues**

- Improve service-level catalogues
    - Cover all the datasets provided by the service.
    - Provide a minimum set of metadata parameters (agreed at Copernicus level).
    - Provide a unique dataset identifier (dataset_id).
    - Provide one PUM for each dataset.
- Develop a Copernicus-level catalogue that builds on the existing service-level catalogues.

**On data formats & data dissemination**

- Implementation of standards (format, structure) will facilitate interoperability of the service and users.
- Move portfolios to cloud services and make data and metadata accessible through APIs.

# 5 Conclusions

In this study we present an AI-based methodology to identify and classify quality layers of Copernicus services. As a pilot case, we applied the methodology to the Copernicus Land Monitoring Service (CLMS) portfolio composed by 94 datasets and 757 layers. We identified 189 quality layers across 61 datasets; 14 datasets reported uncertainties, but only 8 datasets followed metrological guidelines: global NDVI at 300m, global Lake Surface Water Temperature at 1km, global land surface temperature at 5km, global Sentinel-2 mosaic variables derived using SIAC, and some of the European cropland layers. These could support use cases demonstrating how propagating uncertainties to policy indicators can enhance the uptake of Copernicus data in policymaking. Note that all of them make assumptions on the uncertainty sources considered and their characterization, but they are good steps towards the right direction. To improve, a better definition of the uncertainty budgets in the documentation and the provision of coverage factors is needed.

Regarding quality indicators in general, we found inconsistent quality nomenclature in CLMS, with terms such as uncertainty, error, and standard deviation used interchangeably. Also, generic layer names such like flag or confidence varying in meaning across datasets. CLMS, and Copernicus in general, should work towards standardization of content and terminology of quality layers.

The main barrier to analyse CLMS quality layers was the lack of programmatic metadata access. To extend this study to all Copernicus services, the development of comprehensive service-level metadata catalogues built on standards and reporting at a minimum set of metadata parameters.

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## List of abbreviations and definitions

| Abbreviations | Definitions |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| ATBD | Algorithm Theoretical Basis Document |
| BDAP | Big Data Analysis Platform |
| BIPM | Bureau International des Poids et Mesures |
| BRF | Bidirectional Reflectance Factor |
| C3S | Copernicus Climate Change Service |
| CAMS | Copernicus Atmosphere Monitoring Service |
| CCI | Climate Change Initiative |
| CDSE | Copernicus Data Space Ecosystem |
| CEMS | Copernicus Emergency Monitoring Service |
| CEOS | Committee on Earth Observation Satellites |
| CLMS | Copernicus Land Monitoring Service |
| CMEMS | Copernicus Marine Environment Monitoring Service |
| CSV | Comma Separated Values |
| EEA | European Environment Agency |
| EGMS | European Ground Motion Service |
| EO | Earth Observation |
| ESA | European Space Agency |
| ESRI | Environmental Systems Research Institute |
| EU | European Union |

| Abbreviations | Definitions |
|---|---|
| FAO | Food and Agriculture Organization |
| fAPAR | Fraction of absorbed photosynthetically active radiation |
| FCOVER | Fraction of Vegetation Cover |
| GBOV | Global Ground-based Observations for Validation |
| GeoTIFF | Geospatial Tag Image File Format |
| GUM | Guide of the Expression of Uncertainty in Measurement |
| HRL | High Resolution Layer |
| HRP_VPP | High-Resolution Vegetation Phenology and Productivity |
| HSM | Hot Spots Monitoring |
| IDE | Integrated Development Environment |
| JCGM | Joint Committee for Guides in Metrology |
| JRC | Joint Research Centre |
| JSON | JavaScript Object Notation |
| LAI | Leaf Area Index |
| LIE | Lake Ice Extent |
| LST | Land Surface Temperature |
| LWQ | Lake Water Quality |
| LSWT | Lake Surface Water Temperature |
| NBAR | Normalized Bidirectional Reflectance |
| NDVI | Normalized Difference Vegetation Index |
| NetCDF | Network Common Data Form |
| OLCI | Ocean and Land Colour Instrument |

| Abbreviations | Definitions |
|---|---|
| OPT MPC | Optical Mission Performance Cluster |
| PUM | Product User Manual |
| QA4EO | Quality Assurance Framework for Earth Observation |
| RMSE | Root Mean Square Error |
| S2GM | Sentinel-2 Global Mosaic |
| SIAC | Sensor Invariant Atmospheric Correction |
| SLSTR | Sea and Land Surface Temperature Radiometer |
| SSM | Surface Soil Moisture |
| SSMIS | Special Sensor Microwave Imager/Sounder |
| STAC | Spatio-Temporal Asset Catalogue |
| SWE | Snow Water Equivalent |
| SWI | Soil Water Index |
| TIFF | Tag Image File Format |
| TOC-r | Top of Canopy normalized reflectance |
| VIMs | International Vocabulary of Metrology |
| WL | Water Level |

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